

CAREER PROFILE

Senior AI and Data Science Engineer with over 15 years of experience spanning across diverse IT domains.

I specialize in building intelligent systems that unify classical statistical modeling with modern LLM ecosystems, enabling adaptive, explainable, and automated decision workflows.

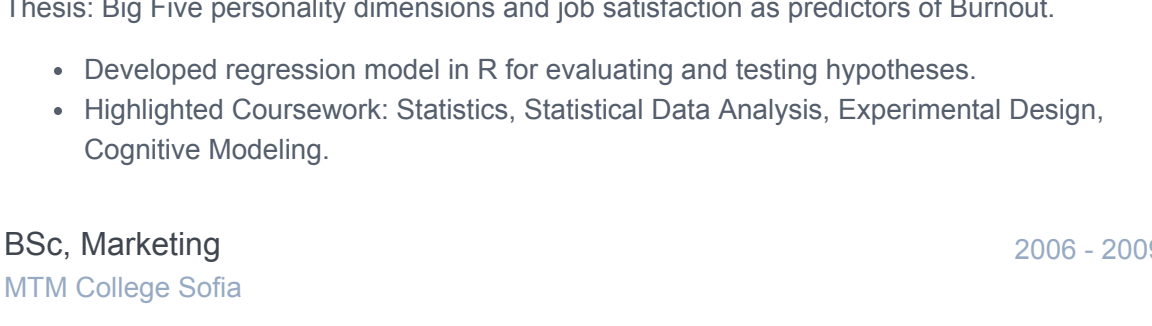
My recent work focuses on architecting multi-agent platforms delivering enterprise-grade AI services.

I combine deep expertise in Python, distributed computing, and MLOps to design reliable, observable, and cloud-native solutions.

Having led cross-functional data teams, I excel in translating complex organizational needs into scalable architectures that balance innovation, reliability, and responsible AI governance.

Passionate about advancing practical AI, I continuously evaluate emerging tools and frameworks to deliver production-ready, secure, and high-impact intelligent systems.

SKILLS & PROFICIENCY



EDUCATION

MSc in Cognitive Science 2022 - 2024
New Bulgarian University

Delving into the intricacies of human cognition and the underlying neural mechanisms. Specializing in Neuroscience, Cognitive Modeling, and Experimental Design.

- Conducting advanced research emphasizing on cognitive control and working memory.
- Utilizing electrophysiological methods to capture and analyze neural responses during cognitive tasks.
- Employing Python to process, analyze, and derive meaningful insights from complex, multivariate neuroscientific data.

BSc in Psychology 2016 - 2020
New Bulgarian University

Thesis: Big Five personality dimensions and job satisfaction as predictors of Burnout.

- Developed regression model in R for evaluating and testing hypotheses.
- Highlighted Coursework: Statistics, Statistical Data Analysis, Experimental Design, Cognitive Modeling.

BSc, Marketing 2006 - 2009
MTM College Sofia

Highlighted Coursework: Statistics, Higher Maths, Finance, Data Analysis, Marketing Research.

EXPERIENCE

Senior Data Science Engineer 2024 - Current
Mach6 Technology

Steering AI system architecture and applied research initiatives within the Governance, Risk Management, and Compliance (GRC) sector.

- Defining the company's generative AI strategy and delivering production-grade agentic platforms at scale.
- Designing and standardizing internal agentic architecture spanning orchestration, retrieval-augmented generation (RAG), and evaluation frameworks used across multiple enterprise products.
- Leading end-to-end integration of emerging frameworks into unified pipelines supporting adaptive reasoning, hybrid search, and autonomous workflow execution.
- Establishing operational standards for observability, testing, and CI/CD, embedding evaluation metrics, telemetry, and reproducible containerization workflows across deployments.
- Partnering with product leadership to translate compliance and risk-management requirements into deployable AI capabilities for automation and decision support.

Senior Data Science Engineer 2022 - 2024
Prime Holding JSC

Led a team of 5 data engineers and scientists, harnessing LLMs and a diverse open-source stack to create novel software.

- Pioneered the application of vector databases in conjunction with LLM based agents for retrieval-augmented generation and information retrieval.
- Fine-tuned foundational LLMs with domain specific datasets using quantized low-rank adaptation methods (qLORA).
- Engaged in presales activities and worked closely with prospective clients to help define their requirements and align them with tailored solutions.
- Created detailed designs and architectural diagrams that encapsulate data strategy and methodology, ensuring a clear roadmap that is well-aligned with client objectives.

Data Science & Analytics Consultant 2020 - 2021
Hitachi Solutions Europe

Worked as a full stack data scientist on variety of projects.

- Established pan-European practice driving the expansion of data science portfolio.
- Participated in large-scale data migration and augmentation projects using Azure Data Factory, Synapse, and Power BI integration.
- Used R-Shiny and Dash to integrate R and Python into containerized predictive web applications deployed to the Azure cloud.
- Used Azure DevOps for backlog and version (Git) control.
- Used Docker for application isolation and deployment.

Business Intelligence Developer 2020 - 2020
Hewlett Packard Enterprise

Collaborated with the business to maintain and update critical company tools, ensuring the proper use of KPIs and metrics. Updated and automated SQL Views and SSIS packages used for data loads of millions or records.

- Participated in the development of real-time streaming analytics solution for monitoring of Salesforce cases, orders, and escalations. Used Azure SQL DB to store live data feed from Kafka consumer, connected with direct query to a Power BI dashboard presenting instant view of the data.
- Developed end-to-end machine learning algorithm written in R to forecast demand for one of HPE's GreenLake strategic deals. Used combination of Kwiatkowski-Phillips-Schmidt-Shin (KPSS) and augmented Dickey-Fuller (ADF) tests to evaluate hypotheses and determine time series stationarity.
- Developed Salesforce connector with R for on demand extraction of objects and fields using complex SOQL queries and API calls (REST & BULK) wrapped in functions, thus automating traditional salesforce reporting and data extraction for large volumes of data (+30 million rows).

Business Analyst 2018 - 2020
DXC Technology

Analyzed organizational and business domain specific processes and systems, assessed the business model and its integration with technology within the boundary of Workplace and Mobility of Offering Delivery & Transformation.

- Developed Power BI dashboards to monitor headcount attrition and turnover, employee distribution, skillset, and active project assignments.
- Worked with global L5 management team to drive business decisions, leveraging data from Salesforce, SAP Business Objects, and other internal systems.

Solution Consultant 2016 - 2018
Hewlett Packard Enterprise

Identified and validated opportunities to accelerate consumption in high potential customer accounts. Drove solution architecture in partnership with the delivery teams. Enabled the customer to envision their environment and services, while leveraging the greater DXC and partner resources (internal teams and partners).

- Provided technology leadership for development of Future Mode of Operation solution designs and functional migration plans. Closely cooperated with Account Teams, Subject Matter Experts, Chief Engineers, Operations, Delivery and Clients.
- Worked with financial data during pre-sales. Created cost models and revenue plans.

PUBLICATIONS

- OKR Platform, Powerful Machine Learning Always on Target.**
Metodi Simeonov
July 2022

PROJECTS

Selected projects from my work, each highlighting a distinct system or capability built as part of a broader AI and data platform strategy.

MCP Agent Platform (July 2025 - Current) - Architecting enterprise-grade FastAPI service exposing LLM-driven MCP agents augmented with static/remote tooling, semantic routing, and hybrid integrations across Salesforce, Commercetools, a CRM, and other systems for customer operations automation. Designed streaming SSE adapters, semaphore-guarded Autogen runtimes with JWT propagation, and Redis-backed context hydration to support resilient multi-session workflows. Extended Autogen's MCP workbench to annotate JSON-schema tools, auto-recover, and emit AG-UI events powering rich, observable UI feedback loops. Automated builds and quality gates with Makefile, Docker, pre-commit hooks, GitHub Actions, Trivy image scanning, and Snyk security policy enforcement.

Diagnostics & Recommendation Engine (August 2025 - Current) - Building an asynchronous diagnostics backend that consumes Azure Service Bus events, validates complex payloads, and performs single-shot LLM-driven analysis via AutoGen for deterministic, Pydantic-validated outputs containing analysis of risk and product recommendations. Engineered a lifespan-controlled FastAPI service with semaphore-based backpressure, auto-reconnect logic, and explicit acknowledge semantics, ensuring reliable, concurrent message handling. Encapsulated Service Bus messaging through a broker abstraction, templated prompts, and centralized telemetry with Azure/MLMP filtering. Enforced robust quality gate through async pytest coverage of concurrency and broker workflows, Makefile pipelines, Dockerized Service Bus Emulator, pre-commit hooks, and Trivy image scanning for production-grade security and compliance.

AI Agent Endpoint (March 2025 - June 2025) - Developed a production-grade multi-agent helpdesk automation service integrating Azure OpenAI, MongoDB Atlas, and SendGrid API. Built an asynchronous FastAPI endpoint coordinating Analysis, Service, and Mail agents via Autogen to interpret queries, retrieve context, and trigger escalation workflows. Implemented hybrid retrieval combining Vector Search and Atlas Search through LangChain and Motor for relevant, low-latency document access. Integrated templated SendGrid notifications with validated environment secrets to ensure secure, deterministic case handoffs. Reinforced reliability with structured logging, health probes, async metrics, and conversation persistence. Delivered an intelligent, self-contained endpoint enabling automatic classification, routing, and escalation of helpdesk requests with consistent, monitored performance.

Forecasting Platform (Jan 2025 - March 2025) - Developed a large-scale forecasting system modeling written premure by country and insurance class using Azure Databricks and Prophet/GLM models. The platform yielded thousands of time-series combinations through permutations of model type, macroeconomic regressors, and cross-validation folds with hyperparameter grid search. Automated model selection and MLflow experiment tracking ensured transparency and reproducibility. Parallelized execution with Python multiprocessing accelerated training and forecasting, while validation logic handled sparse and outlier-prone data. Produced quarterly and yearly forecasts with metadata integrated into Databricks and downstream BI pipelines, delivering reliable, scalable insights for actuarial analysis.

Document Embedding & Hybrid Search Service (Sep 2024 - Dec 2024) - Built a retrieval-augmented generation (RAG) platform delivering grounded enterprise Q&A over internal document collections, powering multiple learning assistants across our GRC portfolio. Designed an ensemble retriever combining MongoDB Atlas Vector Search, BM25, and LLM-based contextual compression through LangChain to balance precision and recall. Implemented evaluation with TruLens for groundedness and answer relevance, storing metrics asynchronously in PostgreSQL. Exposed a FastAPI endpoint that orchestrates Azure OpenAI models with persistent conversation memory, structured logging, and environment-based configuration. Benchmarked performance using a dedicated latency-and-correctness harness on regulatory domain datasets, enabling transparent, production-grade RAG performance monitoring. Shipped the system with Docker including Prometheus and Grafana for observability.

Agent Marking Service (April 2024 - Aug 2024) - Developed and deployed an AI-powered exam marking service that integrates a Streamlit user interface with a FastAPI backend orchestrating Azure OpenAI models. Designed an agentic grading pipeline using Autogen to assemble context from exams, marking guides, and exemplar responses, generating structured grades and feedback stored in MongoDB. Implemented asynchronous background job processing, structured logging, and readiness/health probes for reliable operation. Delivered hardened Docker containers with multi-stage builds, digest-pinned Helm deployments, and environment-sourced secrets managed via Azure Container Registry. Automated builds and Helm chart publishing through GitHub Actions, ensuring full local-to-cloud parity and production-grade delivery.

OCR Automation project (Sept 2023 - Feb 2024) - Architected an automated invoice processing platform combining Azure OCR engines, Adobe REST APIs, and Generative AI for document intelligence. Built Flask-based APIs orchestrating OCR and OpenAI Assistant outputs into unified, validated JSON payloads for enhanced data accuracy. Designed a Python backend handling asynchronous data transformations, structured logging, and reliable ingestion into Supabase. Implemented text-based document classification and integrity checks to support downstream analytics pipelines. Delivered a fully serverless, cloud-native system improving processing efficiency and operational transparency across automation workflows.

RAG Web Application (Jun 2023 - Aug 2023) - Link - Developed a Flask and Streamlit-based application for extracting and interacting with data from vectorized documents using transformer models and langchain. Optimized for performance in Docker with GPU support, the project features integration with pgvector database and efficient model utilization, including Stable-Playtus2-13B and Llama-2-7b-chat-hf. Managed full-stack implementation from data processing and server setup to user interface design, ensuring compatibility with various hardware specifications and deploying effective debugging strategies.

Risk Analysis Application project (Feb 2023 - May 2023) - Link - Designed and developed Postgres deployment and configuration codebase including the underlying data layer of a risk assessment application. Developed multiple custom connectors using Python, supporting dynamic data ingestion from various sources (APIs, local & remote files, etc.). Created consolidated data model through dynamic and materialized views. Fully automated the CI/CD deployment pipeline using Docker and GitHub Workflows. Achieved complete database and data-layer integration as a microservice within a Kubernetes cluster.

Anomaly Detection Application project (Oct 2022 - Feb 2023) - Deployed project infrastructure using Docker, Postgres, Prometheus, Grafana and MLFlow. Researched anomaly detection methods based on confidence and probability intervals with detection of anomalous groups (Entropy, Wavelet transforms, Isolation Forest, Prophet).

Kafka-Streaming project (Oct 2022 - Feb 2023) - Link - Developed Kafka data-streaming solution. Designed and implemented a real-time data pipeline. Integrated Spark Streaming with Kafka for data ingestion. Applied data cleaning and sentiment analysis using TextBlob. Stored processed data in Atlas MongoDB cloud.

Smart Alignment project (Aug 2022 - Oct 2022) - Developed preliminary dataset and analyzed semantic similarity of key product features using sentence transformer model (SBERT), thus improving KPI automation process.

Customer churn analysis case study (July 2022 - Aug 2022) - Created end-to-end project design and implementation. Developed novel data model sourcing data from 5 systems (Redis, Chargebee, Salesforce, MongoDB, Postgres) using Azure Synapse. Iterated through multiple classification models and developed R-Shiny web application for interactive customer churn scoring.

Automatic Insights Recommendation project (May 2022 - Aug 2022) - Developed TensorFlow multiclass classification model with hyperparameter optimization. Integrated model tracking and testing with MLFlow.

Autofill OKRs project (Jan 2022 - Mar 2022) - Developed language detection inference pipeline, used LIME for explainability of a large TensorFlow multiclass classification model. Developed complex data model sourcing data off production systems using Azure Synapse, AZDL and PGDB. Used Grafana to develop custom metrics and KPIS for tracking model performance in production.

Equity Trading Forecast Analysis (Dec 2021) - Link - Worked on a POC consultancy project for a FinTech company where I developed a forecasting solution for predicting equity trading volumes. Used variety of nested and ensembled models in R to process multiple time series in parallel.

Nested Forecaster Web App (Oct 2021) - Link - Developed R-Shiny Product Forecasting App for the M5 Forecasting Accuracy challenge on Kaggle using XGBoost, Prophet and Random Forest models.

Hierarchical Forecaster Web App (Sept 2021) - Link - Developed R-Shiny Product Exploration App for the M5 Forecasting Accuracy challenge on Kaggle using an ensemble approach comprising of XGBoost and Lightgbm models.

Employee Attrition Analysis (July 2021) - Link - Developed predictive analysis to target employee attrition. Used variety of tree based and ensemble models in H2O, embedded in Azure Machine Learning.

CERTIFICATIONS

Agentic AI 2025
DeepLearning.ai

Reasoning with o1 2025
DeepLearning.ai

Automated Testing for LLMOps 2024
DeepLearning.ai

LLMops 2024
DeepLearning.ai

Understanding and Applying Text Embeddings 2024
DeepLearning.ai

Building and Evaluating Advanced RAG 2023
DeepLearning.ai

Finetuning Large Language Models 2023
DeepLearning.ai

LangChain for LLM Application Development 2023
DeepLearning.ai

Vector Databases from Embeddings to Applications 2023
DeepLearning.ai

ChatGPT Prompt Engineering for Developers 2023
DeepLearning.ai

Python Data Structures 2023
Udemy

AWS Cloud Technical Essentials 2022
Coursera

AWS Fundamentals 2022
Udemy

DS4B 101-P Python for Data Science Automation 2022
Business Science University

DS4B 201-R Data Science For Business With R 2021
Business Science University

DS4B 203-R High Performance Time Series Forecasting 2021
Business Science University

DS4B 102-R Web Applications with R-Shiny 2021
Business Science University

DP-900 Microsoft Azure Data Fundamentals 2021
Microsoft

AZ-900 Azure Fundamentals 2021
Microsoft

DS4B 101-R R for business analysis 2021
Business Science University

Distributed Computing with Spark SQL 2020
Coursera

Data Wrangling, Analysis and AB Testing with SQL 2020
Coursera

Advanced R Programming 2020
Coursera

Building Interactive Apps With Shiny and R 2020
Udemy

Programming with R 2020
Coursera

Qlik Sense Data Architect Masterclass 2020
Udemy

MCSA Machine Learning 2019
Microsoft

773 Analyzing Big Data with Microsoft R 2019
Microsoft

774 Perform In Cloud Data Science with Azure Machine Learning 2019
Microsoft

SQL for Data Science 2019
Coursera

Implementing a Data Warehouse with Microsoft SQL Server 2019
Udemy

Perform ETL using SSIS 2019
Udemy



Metodi Simeonov

Senior Data Science Engineer

✉ metodi.simeonov@gmail.com

☎ +359 887 544 993

🌐 metodi.github.io/cv/

📧 mesime

🔗 metodi

LANGUAGES

Bulgarian (Native)

English (C1)